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Endogenous innovation and imitation in a model of equilibrium growth

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Abstract

This paper provides a growth model in which innovation as well as imitation occurs. Economic growth is due to product innovations. Innovators driven by the possibility to appropriate monopoly profits do not remain in their monopolistic position forever. Latecoming imitators get into possession of the private knowledge of production through investments in R & D. Imitated products are marketed in oligopolistic markets. Imitation proves to be profitable despite a single factor market and positive imitation costs. A steady-state equilibrium with positive imitation and innovation rates as well as different market structures can be derived. Finally, the effects of industrial policy measures are discussed.

JEL classification: O31; O38; O40

Keywords: Innovation; Imitation; Growth

1. Introduction

Product and process innovation is a major, if not *the* driving force of economic growth. Through the development of new products and processes, an innovator achieves a temporary advantage earning temporary monopoly profits. But, often the innovator loses the monopolistic position through imitation. After imitation

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oligopolistic competition prevails. In this paper, we will draw attention to this specific innovator–imitator relationship in a dynamic general equilibrium model. The model is based on the endogenous growth approach of Romer (1990) and Grossman and Helpman (1991a). Endogenous growth stems from development of new intermediate goods leading to productivity growth in the high-technology final good using the intermediate products as inputs.

The existing literature mainly concentrates on innovation. The R&D sector is characterized by either the assumption of perfect patent protection (cf. Romer, 1990) or by the assumption of Bertrand–Nash strategies used by the innovators in the differentiated goods production (cf. Grossman and Helpman, 1991a). If imitation actually occurs in these models, imitation is either costless (cf. Dinopoulos et al., 1993) or producers face different factor costs (cf. Grossman and Helpman, 1991b). Both assumptions (perfect patent protection and Bertrand–Nash behaviour) can not considered being very realistic, at least not in all situations. On the one hand, patent protection in many industries is not used in most cases and is not always considered the most efficient means of protection against imitation (cf. von Hippel, 1988), and patents are also not able to protect against imitation in all cases (cf. Mansfield et al., 1981). On the other hand, the assumption of Bertrand–Nash behaviour, implying marginal cost pricing, sets forth an 'extreme model' (Tirole, 1990, p. 402). The use of a limit-price strategy to deter entry works only in a price setting game. In the case of quantity setting it represents a noncredible threat.

In addition, these approaches make no allowance for the empirical fact that imitation of product innovations, despite positive imitation costs is a rather commonly observed phenomenon (cf. Levin et al., 1987), often taking place in the same economy or between economies with similar factor prices. We use an alternative modelling approach for the post-entry game in the intermediate goods sector by assuming quantity-setting oligopolists. Duopolistic profits are lower than monopolistic ones, but might very well be sufficient anyway letting imitation be profitable. Hence, some of the intermediate goods are produced by more than one supplier. In contrast to existing approaches, in this paper, monopoly pricing does not prevail in all market segments of the intermediate goods sector. Monopolistic and duopolistic market structures coexist. There is a constant change in market structure. For whereas monopolistic markets are newly created through the development of new goods other market segments change from a monopoly to a duopoly state due to imitation.

We show that a steady-state equilibrium with positive and constant innovation and imitation rates exists. Our model is related to the approach of Segerstrom (1991). However, he analyzes 'vertical' product innovations with quality improvements. Furthermore, he models duopolistic interaction as a supergame with a cooperative solution, i.e. in his model the monopoly price is charged even after imitation has occurred. Some of his results substantially differ from the ones derived in this paper. Another difference is that we analyze the behavior of the economy out of steady state. We also get different results compared to the North–South model of Grossman and Helpman (1991b).

The presented model is distinctively different to the older approaches on imitation and diffusion in growth models (cf. e.g. Nelson and Phelps, 1966; Nordhaus, 1969). Whereas in the older literature, imitation and diffusion are exogenous and costless processes, here, imitation is endogenously determined, based on the decisions of forward-looking firms taking the imitation costs into account. In this paper, imitation is meant to be the copying of new knowledge, rather than the adoption process of the new technology. In contrast to the industrial organization literature (cf. e.g. Tirole, 1990, chapter 10), in the present work the imitation process is embedded in a dynamic general equilibrium model.

The paper is organized as follows. In Section 2 we derive the basic model. In Section 3 we discuss the existence of a steady-state equilibrium with positive and constant rates of innovation and imitation. Industrial policy measures on the steady-state equilibrium are considered in Section 4. The last section concludes.

2. The model

We consider an economy with two final goods. Every consumer inelastically supplies one unit of labor and possesses the intertemporal utility function at time t:

$$U_t = \int_t^{\infty} e^{-\theta(\nu - t)}\{\sigma \ln C_Y(\nu) + (1-\sigma)\ln C_Z(\nu)\}\mathrm{d}\nu, \quad \sigma < 1. \tag{1}$$

The rate of individual time preference is represented by θ. Consumption of the industrial (traditional) good is denoted by C_Y (C_Z). Taking the intertemporal budget constraint into account we get for the optimal consumption path the Euler equation: [1]

$$\dot{E}/E = r(t) - \theta. \tag{2}$$

Given that the market rate of interest $(r(t))$ at time t just equals the subjective rate of time preference it proves optimal to leave consumption expenditures (E) constant over time. With regard to the simple static maximization problem, we find

$$\sigma E = p_Y C_y \tag{3a}$$

and

$$(1-\sigma)E = p_Z C_Z, \tag{3b}$$

where the p_l $(l = Z, Y)$ stand for the respective consumer prices.

[1] In the following, dots (hats) represent time (growth) rates.

One of the main features of the analysis is the production function of the Y good:

$$Y = L_Y^{1-\beta}\left[\int_0^n x(j)^{\alpha}\mathrm{d}j\right]^{\beta/\alpha}; \quad 0<\alpha,\beta<1, \tag{4}$$

where L_Y stands for the labor input in Y, and $x(j)$ represents the amount of the jth intermediate good. Let n denote the number of known intermediate goods. Since Y-producers take n as given, (4) is compatible with perfect competition in the Y-sector. Profit-maximization of the Y-producers yields the demand functions for every intermediate good j' (cf. Helpman and Krugman, 1985):

$$x(j') = \frac{p_x(j')^{-\epsilon}}{\int_0^n p_x(j)^{1-\epsilon}\mathrm{d}j}\beta p_y Y, \tag{5}$$

with $p_x(j)$ representing the price of the jth intermediate product and $\epsilon = 1/(1-\alpha)$. The intermediate goods are assembled with labor alone:

$$x(j') = \frac{L_x(j')}{a_x}, \tag{6}$$

where a_x depicts the labor input coefficient in the x-production and L_x is the demand of labor of this sector. Every intermediate goods producer chooses $x(j)$ to maximize

$$G_x(j) = \{p_x(j) - a_x w\}x(j), \tag{7}$$

with w as the wage rate. We distinguish two types of intermediate-goods suppliers. In n_{m} markets, the innovator holds a monopoly position, the product is not imitated yet. There is one innovator and one imitator each in n_{d} ($n_{\mathrm{d}} = n - n_{\mathrm{m}}$) markets. The duopolists use Cournot–Nash strategies. [2] For the monopoly price, we get by maximizing (7) and using (5)

$$p^{\mathrm{m}} = \frac{a_x w}{\alpha}. \tag{8}$$

[2] The use of Cournot strategies can be rationalized on various grounds. First, with quantity-setting, the limit-price of the incumbent (or the respective output) represents a noncredible threat to the newcomer (cf. Friedman, 1983). By anticipating the post-entry game structure correctly, the newcomer will realize that in the post-entry game a Nash–Cournot equilibrium will result and that the limit-price does not constitute profit-maximizing behavior of the incumbent after market entry has occurred. Entry can not be deterred by this measure. Furthermore, the Cournot–Nash strategy avoids the rather unrealistic feature of Bertrand–Nash competition in homogeneous goods markets: that the competition of only two firms leads to marginal cost pricing. An additional, rather convincing argument for the Cournot solution was brought forward by Kreps and Scheinkmann (1983). They showed that the Cournot outcome can be interpreted as the result of a two-stage game, in which in a first stage capacities are chosen and in a second stage a price-setting game takes place.

In the market segment with already imitated goods, duopolists face the same decision problem and the same cost functions. We get as the optimal duopolistic price

$$p^{\mathrm{d}} = \frac{2a_x w}{1+\alpha}. \tag{9}$$

By using (5), (7), (8) and (9), the profit function of the n_{m} producers is

$$G^{\mathrm{m}} = \frac{(1-\alpha)\delta\beta p_Y Y}{n_{\mathrm{m}}\delta + n_{\mathrm{d}}}, \tag{10}$$

with $\delta = (2\alpha/(1+\alpha))^{\alpha/(1-\alpha)} < 1$. Each of the $2n_{\mathrm{d}}$ duopolists achieves a profit level of

$$G^{\mathrm{d}} = \frac{\beta p_Y Y(1-\alpha)}{4(n_{\mathrm{m}}\delta + n_{\mathrm{d}})}. \tag{11}$$

The blueprints of products in the intermediate-goods sector can be newly developed or copied by investing in the R&D sector. Borrowing from Romer (1990) and Grossman and Helpman (1991a) we assume that the R&D sector creates public as well as private knowledge. Innovators and imitators, respectively, face one of the two functions for the development or imitation of an additional intermediate good:

$$\dot{n} = \frac{WL_n^{\mathrm{in}}}{a_{\mathrm{in}}}, \tag{12a}$$

$$\dot{n}_{\mathrm{d}} = \frac{WL_n^{\mathrm{im}}}{a_{\mathrm{im}}} \tag{12b}$$

where $L_n^{\mathrm{in}}(L_n^{\mathrm{im}})$ delineates the labor input of innovators (imitators). The level of public knowledge, W, contributes positively to the productivity in the R&D sector. In our model of a closed economy, it is rather plausible to assume that imitators and innovators have access to the same pool of public knowledge. We assume the level of public knowledge being proportional to the number of developed products. For simplicity let $W = n$. The terms a_{in} and a_{im}, respectively, depict the input coefficients of labor in the innovation and imitation processes, with $a_{\mathrm{in}} > a_{\mathrm{im}}$. This relation is in accord with empirical findings (see Mansfield et al., 1981). Hence, $c_n^{\mathrm{in}} = a_{\mathrm{in}} w/n > c_n^{\mathrm{im}} = a_{\mathrm{im}} w/n$. Innovation costs are larger than the costs of imitating.

There is free entry in both R&D activities. Hence, in an equilibrium with positive R&D, discounted future benefits of innovation (imitation) activities ν_{in} (ν_{im}) are just equal to the respective R&D costs. For the imitation sector this is

$$v_{\mathrm{im}} = \int_t^{\infty} \mathrm{e}^{-(R(\nu)-R(t))} G^{\mathrm{d}}(\nu)\,\mathrm{d}\nu = \frac{a_{\mathrm{im}} w}{n}, \tag{13}$$

with $R(t)$ being the accumulated rate of interest at t. Differentiation of (13) with regard to t results in the no-arbitrage condition for imitation:

$$G^{\mathrm{d}} + \dot{v}_{\mathrm{im}} = rv_{\mathrm{im}}. \tag{14}$$

Imitators are targeting market segments with a monopolistic structure since profits are lower in a market with three firms compared to duopolistic profits.

As to the no-arbitrage condition of innovators, the main difference can be seen in the fact that the innovator temporarily receives monopoly profits but experiences the threat of a partial capital loss with probability $\dot{n}_{\mathrm{d}}/n_{\mathrm{m}}$, the rate of imitation m. In this case the firm's value reduces from v_{in}, the expected discounted value of the innovator to v_{im}. The no-arbitrage condition of innovation is therefore: [3]

$$G^{\mathrm{m}} - m\{v_{\mathrm{in}} - v_{\mathrm{im}}\} + \dot{v}_{\mathrm{in}} = rv_{\mathrm{in}} = r \cdot \left(\frac{a_{\mathrm{in}} w}{n}\right). \tag{15}$$

The traditional good is produced with labor (L_Z) only:

$$Z = L_Z. \tag{16}$$

Commodity market clearance requires the matching of amounts supplied and demanded in both final goods markets, i.e. $C_Y = Y$ and $C_Z = Z$. From (4), (12), and (16), as well as (6), (8), (9), and (5) for the intermediate goods sector, the labor market clearing condition $L = L_n^{\mathrm{in}} + L_n^{\mathrm{im}} + L_x + L_Y + L_Z$ can be written as

$$L = a_{\mathrm{in}}\frac{\dot{n}}{n} + a_{\mathrm{im}}\frac{\dot{n}_{\mathrm{d}}}{n} + \frac{(1-\beta)\sigma E}{w} + \frac{(1-\sigma)E}{w} + \frac{\beta\sigma E}{2w(n_{\mathrm{m}}\delta + n_{\mathrm{d}})} \times \{n_{\mathrm{d}}(1+\alpha) + 2\alpha\delta n_{\mathrm{m}}\}. \tag{17}$$

Eq. (17) illustrates an important feature of our model and contrasts substantially with innovation–imitation approaches in the presence of international trade (e.g. Dinopoulos et al., 1993; Grossman and Helpman, 1991b). In our model both R&D-activities are tied up with each other through the full employment condition (17). Finally, we normalize prices such that E is equal to one.

3. The steady-state equilibrium

In this chapter the conditions for a unique steady-state equilibrium with positive and constant rates of imitation and innovation as well as with a constant intersectoral factor allocation will be derived. Eq. (17) illustrates that this requires a constant wage rate in time and a constant rate of innovation $\dot{n}/n = g$, too. In addition, in the long-run equilibrium with positive labor demand in each sector,

[3] For a similar derivation see Grossman and Helpman (1991a, p. 288).

the relation between the different market types n_{m} and n_{d} has to remain constant. This implies that the growth rates of n_{m} and n_{d} have to be equal and correspond to the innovation rate g. Hence,

$$\frac{n_{\mathrm{m}}}{n} = \frac{g}{g+m} \quad \text{and} \quad \frac{n_{\mathrm{d}}}{n} = \frac{m}{g+m}.$$

The growth rate of static utility ($\hat{V}$) is proportional to the rate of innovation. As $\hat{Z} = 0$, it follows that $\hat{V} = \sigma \hat{Y}$. From (4), we get as a result of $X = n_{\mathrm{m}} x_{\mathrm{m}} + n_{\mathrm{d}} x_{\mathrm{d}}$ being constant in a steady-state equilibrium, $\hat{Y} = [\beta(1-\alpha)/\alpha]g$, and $\hat{V} = [\sigma\beta(1-\alpha)/\alpha]g$.

The two no-arbitrage conditions as well as the factor market clearance condition are the decisive equations to determine the steady-state equilibrium. Taking the steady-state conditions as well as (2), (3), (10), (11), (12), and the market-clearing conditions into account, these three equations can be written as

$$\frac{\beta\sigma(1-\alpha)(g+m)}{4a_{\mathrm{im}} w(g\delta+m)} - g = \theta, \tag{14'}$$

$$\frac{\beta\sigma\delta(1-\alpha)(g+m)}{a_{\mathrm{in}} w(g\delta+m)} - m\left(1 - \frac{a_{\mathrm{im}}}{a_{\mathrm{in}}}\right) - g = \theta, \tag{15'}$$

$$L = a_{\mathrm{in}} g + \frac{a_{\mathrm{im}} gm}{g+m} + \frac{1}{w}\left(1 - \beta\sigma(1-\alpha) + \frac{\beta\sigma(1-\alpha)}{2}\frac{m}{m+\delta g}\right). \tag{17'}$$

These three steady-state equations can be reduced to two remaining equations. Solving (14′) for w, and using the result in (15′) yields

$$m^{\mathrm{CI}} = \frac{4\delta(a_{\mathrm{im}}/a_{\mathrm{in}}) - 1}{1 - a_{\mathrm{im}}/a_{\mathrm{in}}}(g+\theta). \tag{18}$$

Thus m^{CI} represents the rate of imitation that for alternate rates of innovation is compatible with a steady-state equilibrium in the innovation as well as the imitation area. In the following, we impose the following parameter restriction: $a_{\mathrm{im}} > ((\delta+1)/5\delta)a_{\mathrm{in}}$. [4] On the one hand this restriction proves to be a sufficient condition for the stability of our equilibrium. On the other hand it allows for a simultaneous equilibrium in the innovation and imitation sector. It guarantees that profits per share in the innovation sector are larger than the ones in the imitation sector, i.e. $G^{\mathrm{m}}/v_{\mathrm{in}} > G^{\mathrm{d}}/v_{\mathrm{im}}$ (see (10), (11), (13), and (15)). Shareholders are willing to hold shares of innovating as well as imitating firms if the respective profits per share exceed the risk-free interest rate ($r = \theta$) by the amount of g (see (14′) and (15′)). But in addition innovators' profits per share have to compensate

[4] Note that δ is decreasing in α and $\lim_{\alpha \to 1} = e^{-0.5} = 0.6065$.

for partial capital losses by means of imitation $(-m(1-a_{\text{im}}/a_{\text{in}}))$. Given our assumption, a simultaneous equilibrium in both R&D sectors can be sustained. m^{CI} is upward sloping. The positive slope can be explained as follows. Suppose, starting on m^{CI} that g increases. This leads to an increase in profits per share of imitating firms as well as to a larger expected capital loss in the imitation sector. The first effect overcompensates the latter as can be seen from (14′) and (15′). Hence, entry in the imitation sector takes place. The rise in the imitation rate lowers profits per share of an imitating firm. It also increases the expected capital loss for the innovators. Therefore, the increase in g leads to a larger m until both no-arbitrage conditions are fulfilled again.

Inserting (14′) into (17′) yields

$$m^{\text{CL}} = \frac{(1-\alpha)(L - a_{\text{in}}g) - 2a_{\text{im}}\tau(g+\theta)}{(1-\alpha)(a_{\text{in}} + a_{\text{im}}) + 2a_{\text{im}}\gamma + [2a_{\text{im}}\theta\gamma - (1-\alpha)L]/g}, \tag{19}$$

with $\tau = 2\delta\{(\sigma\beta)^{-1} - (1-\alpha)\}$, $\gamma = 2(\sigma\beta)^{-1} - (1-\alpha)$, and $\gamma > \tau > 0$. $m^{\text{CL}}(g)$ represents the partial steady-state loci in the labor market and the imitation sector.

A steady-state equilibrium emerges at the intersection of (18) and (19). The condition $L > 2a_{\text{im}}\gamma\theta(1-\alpha)^{-1}$ is sufficient for the existence of a unique steady state with positive rates of innovation and imitation [see Appendix A]. The intuition behind this restriction is the following. This condition ensures that the economy has a large enough labor force leading to a factor market clearing wage rate that allows for R&D costs being sufficiently low to make investment in both R&D-sectors profitable. Looking at (14′), (15′), and (17′) reveals that with this condition, $m = g = 0$ does not constitute a long-run equilibrium. The condition expresses the fact that lenders must be sufficiently patient (low θ) to make investments in R&D in return for future profits worthwhile. In addition, it guarantees that market shares $(\sigma\beta)$ and profits in the intermediate goods sectors $(1-\alpha)$ are not too small to leave R&D investment unprofitable. Put together, this condition represents intuitively interpretable parameter restrictions in order to guarantee profitability in the R&D sector. The steady-state is a stable saddle point (see Appendix A).

Given the above condition, m^{CL} represents a downward sloping line in the positive orthant. The intuition behind this is the following. Suppose a larger m than at the initial point on m^{CL}. This leads to a higher demand for labor in the imitation and the intermediate-goods sector. Labor is shifted away from the innovation sector, i.e. g decreases. The reduction in g implies a decrease (increase) of labor demand in the R&D (intermediate-goods) sector. Due to our assumption, the first effect dominates the second. Hence, a smaller innovation rate lowers the demand for labor and reestablishes equilibrium in the factor market. The decrease in g as a reaction of the larger m also leads to a new equilibrium in the imitation sector. Fig. 1 portrays the unique steady-state equilibrium.

The trade-off between m and g reflects the fact that imitation and innovation activities demand resources from the same factor market. This can be seen by a

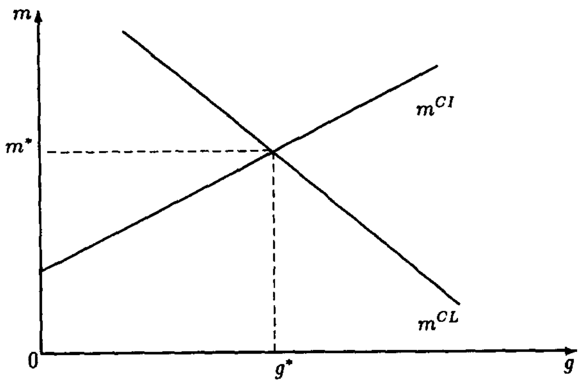

Fig. 1. The steady-state equilibrium.

brief comparison with the North–South model of Grossman and Helpman (1991b). In their two-country model innovation and imitation activities are only linked through the no-arbitrage condition, but not the factor market. The wage rate of the South has no effect on the North. There is a possible positive influence of Northern wages on Southern profits through the pricing condition of the imitators, but not an inverse influence on Southern production costs. In contrast to our approach, none of their equilibrium conditions reflect a trade-off between the rate of imitation and innovation, since there is no direct link between m and g through the factor market.

4. The effects of industrial policy intervention

We will now analyze the effects of small R&D subsidies as well as production subsidies. We assume a balanced discounted budget in the remaining analysis. Then, due to the infinite lifetime of consumers a lump-sum tax used to finance subsidies does not distort the equilibrium allocation.

If cost subsidies (ϕ^{in}) are granted for investments in the innovation area, private costs reduce to $a_{\text{in}}w(1-\phi^{\text{in}})/n$. The subsidies on behalf of innovation costs leave (14′) and (17′) unchanged. Only the no-arbitrage condition in the innovation sector changes. Whereas (19) stays the same, we get for (18):

$$m^{\text{CI}}(\phi^{\text{in}}) = \frac{4\delta\left[a_{\text{im}}/a_{\text{in}}(1-\phi^{\text{in}})\right]-1}{1-\left[a_{\text{im}}/a_{\text{in}}(1-\phi^{\text{in}})\right]}(g+\theta). \tag{20}$$

ϕ^{in} shifts the m^{CI}-curve to the left. In the new equilibrium G in Fig. 2, the rate of imitation is higher and the innovation and growth rates are lower.

Since the imitation rate has an asymmetric effect in the two R&D sectors m has to increase in order to offset the higher profits per share of innovating firms

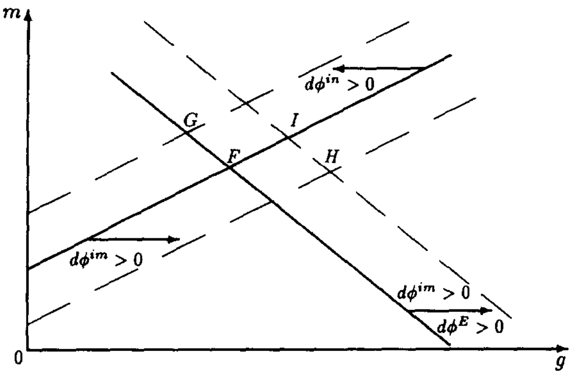

Fig. 2. Effects of state interventions.

which arise through the subsidy. An increase in g would violate the no-arbitrage condition in the imitation sector. A larger m implies a larger demand for labor driving resources away from the innovation sector and lowering g. That is, since m has a larger impact on the innovation sector than in the imitation sector, whereas g has a symmetric one and the two variables are inversely related to each other through the factor market, we get a larger m and a smaller g in the new steady state with an innovation subsidy.

The economic intuition behind this result becomes more clear if we take a look at the transition period and agents' expectations. Agents expect a new interior steady state (with positive rates of innovation and imitation) to emerge. [5] Therefore, it is individually rational to jump on the saddle path being the unique trajectory leading to the new interior solution. For any other trajectory the economy would never reach the new interior steady state thereby violating agents' expectations. The jump on the saddle path implies a higher wage rate in the economy. The increase in the wage rate which eventually overcompensates the innovation subsidy's positive effect on innovators' profits has a stronger negative impact on innovators' expected returns than on the expected return of imitators (see (13)–(15)). In addition, potential innovators foresee a higher rate of imitation. The wage rate effect and the higher imitation rate create a disincentive for

[5] We should mention that there exists another saddle-path stable steady-state equilibrium with a positive innovation and a zero imitation rate (see for the stability of this equilibrium, Appendix B). Since agents live forever, bubbles cannot emerge in our framework. If the subsidy, however, changes agents' expectations and they expect the new equilibrium to be the corner solution with a zero rate of imitation and positive innovation activities, a jump on the saddle path takes place leading to this equilibrium with a larger g. The assumption that all agents believe that an interior solution will emerge after the introduction of an innovation subsidy is especially plausible in the case of small perturbations from the initial steady state with simultaneous innovation and imitation. We concentrate on the perturbation of this equilibrium only.

innovative activities (see (15)) overcompensating the positive effect of the innovation subsidy. Expected returns on innovative activities diminish. Resources are diverted away from the innovation sector and the rate of innovation falls. This, in turn, increases the expected return on imitative activities and leads to a higher rate of imitation in the new steady-state equilibrium.

The influence of the factor market link on our result can be made more obvious with a look at the North–South model of Grossman and Helpman (1991b). In their wide-gap case, in which the Northern wage rate has no direct effect on the no-arbitrage condition of the Southern firms (the imitators), the innovation subsidy has no effect on the innovation rate. In the narrow-gap case Northern wages positively influence profits in the South. In our model we have the reverse relationship: an increase in the innovators' wage rate decreases profits of the imitators. We also get the reverse result. Whereas they find a higher g in the narrow-gap case, we get a lower g.

At a first glance, it seems that our result with respect to the effects of the innovation subsidy parallels the one of Segerstrom (1991). But, whereas here the aggregate rate of innovation and the innovation sector as a whole contract, in Segerstrom's vertical product differentiation model only the innovation rate per targeted market declines. The number of markets targeted by innovators, however, increases as a consequence of the innovation subsidy. Therefore, in his paper, the innovation sector as a whole expands and economic growth is *faster* with an innovation subsidy. This contrasts with our result that the innovation sector in total as well as economic growth declines and represents an important difference.

If subsidies (ϕ^{im}) are exclusively granted to the imitators, only (14′) is changed, (15′) and (17′) stay the same. We get

$$m^{\mathrm{CI}}(\phi^{\mathrm{im}})=\frac{4\delta\left[a_{\mathrm{im}}(1-\phi^{\mathrm{im}})/a_{\mathrm{in}}\right]-1}{1-a_{\mathrm{im}}(1-\phi^{\mathrm{im}})/a_{\mathrm{in}}}(g+\theta), \tag{21}$$

and

$$m^{\mathrm{CL}}(\phi^{\mathrm{im}}) = \frac{g\{(1-\alpha)(L-a_{\mathrm{in}}g)-(1-\phi^{\mathrm{im}})2a_{\mathrm{im}}\tau(g+\theta)\}}{(1-\alpha)(a_{\mathrm{in}}+a_{\mathrm{im}})g+(1-\phi^{\mathrm{im}})2a_{\mathrm{im}}(g+\theta)\gamma-(1-\alpha)L}. \tag{22}$$

Both curves shift to the right. The result is a new equilibrium point (H), with a faster rate of development of new goods and a higher rate of economic growth. The effect on imitation, however, is not clear-cut. The imitation subsidy, which can also be interpreted as a relaxation of patent protection for the innovator, attracts more imitators due to the higher profitability in this sector. But on the other hand, the innovation rate increases in order to reestablish equilibrium in the two R&D sectors. The anticipated increase in g, in turn, leads via the factor market link and higher expected capital losses for imitators to a decline of expected return on imitative activities calling for a smaller m. Depending on which effect dominates m increases or decreases. But, even if the rate of imitation decreases this is not necessarily true for the employment level in the imitation sector $(a_{\text{im}}\, gm/(g+m))$ (see (17′)), since g increases.

If *both* R&D sectors are subsidized a cursory glance at (14′) and (15′) reveals that the effect of this subsidy (ϕ^{E}) on the equations just balances. But (22) shifts to the right and top (see Fig. 2). In the new equilibrium more innovation as well as imitation occurs. With a general subsidy the two no-arbitrage conditions are affected in the same manner, i.e. profits per share increase. Entry in both sectors takes place, i.e. m and g increase. The wage rate effect partially offsets this movement.

We can now also analyze the consequences of production subsidies and look at the effects of ad valorem subsidies and taxes for the individual sectors.

Due to the constant elasticity property of the factor demand functions, the yield per unit for the intermediate-goods producer remains unchanged by means of the production subsidy ϕ^x. However, the input prices of intermediate goods for the Y-producers decrease. The quantity of goods sold by the intermediate-goods producers as well as their profits increase by the factor $(1+\phi^x)$. Taking this into account in (14′), (15′), and (17′), we get precisely the same result as in the preceding section: m^{Cl} remains unchanged. Profit increases in the two sectors just outweigh each other. As a consequence of higher profits in the imitation sector and the increased labor demand in the intermediate-goods sector, Eq. (22), however, reads as

$$m^{\text{CL}} = \frac{g\left\{(1-\alpha)(L-a_{\text{in}}g) - 2a_{\text{im}}(g+\theta)\delta\left(2\alpha+\kappa(1+\phi^x)^{-1}\right\}\right.}{(1-\alpha)\left((a_{\text{in}}+a_{\text{im}})g-L\right)+2a_{\text{im}}(g+\theta)\left((1+\alpha)+\kappa(1+\phi^x)^{-1}\right)}, \tag{23}$$

with $\kappa = (2(1-\sigma)+2(1-\beta)\sigma)/\sigma\beta$. The m^{CL}-curve shifts to the right and the top. We get a new equilibrium with a higher g and m. As a consequence of profit increases in *each* research sector, g and m go up.

The *ad valorem* subsidy on Y-production (ϕ^Y) has the same implications as ϕ^x. The subsidy ϕ^Y increases the demand for Y and hence, for the intermediate goods as well. Profits in both intermediate goods sectors increase by $(1+\phi^Y)$. The above argumentation applies, m and g increase. An *ad valorem* tax on Z-production (ϕ^Z) yields the same qualitative results as the other two production subsidies, but the mechanism differs. With ϕ^Z, (14′) and (15′) are not effected at all. The tax just discriminates against the use of labor in the Z-sector. The workers laid off in the Z-industry are absorbed by the other sectors (m^{CL} shifts outward) leading to a new equilibrium with a higher innovation and imitation rate.

General subsidies in one of the three stages of production in the high-technology sector and taxes discriminating against the Z-sector correspond to each other. Furthermore, our model suggests that from the point of view of a growth-enhancing policy it is better to provide subsidies in the high-technology sector in general rather than to intervene selectively and subsidize only innovative or imitative activities separately.

5. Conclusion

In this paper, a steady-state equilibrium with constant and positive rates of innovation and imitation in the framework of an endogenous growth model was derived. Due to quantity-setting after market entry has taken place imitators can cover their imitation costs by means of future duopolistic profits. One of the main characteristics of our steady-state equilibrium is that it exhibits a permanent change in market structure in the intermediate product sector on the industry-level, whereas on a macroeconomic level the equilibrium relation between the two types of market structure remains unchanged.

In addition to the behaviour of the incumbents after market entry another important feature of our analysis should be stressed. Innovation and imitation demanded labor from the same labor market with a given endowment of labor. Hence, a higher demand for labor in one (R&D) sector leads, ceteris paribus, to a displacement of the other (R&D) sector by way of a higher wage rate. This nexus of the sectors through the labor market was an important factor in our model.

To derive immediate and general policy implications from our simple modelling approach would certainly be overbold. However, we would like to stress the important interaction between innovation and imitation policy makers should take into account.

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Appendix A

Let us define $\mu = n_m/n_d$, the relative number of non-imitated to imitated goods. By noting that $\dot{\mu} = (\mu + 1)(\dot{n}/n - \dot{n}_d/n_d)$ and using (3), (10), (11) as well as the costs of innovation and imitation in (14), (15), and (17) we get after some calculations for the equilibrium dynamics:

$$\dot{w} = \left(\theta + \frac{L}{a_{\text{in}}}\right)w - \frac{\xi(\mu+1)}{4(\mu\delta+1)a_{\text{im}}} - \frac{\mu\xi}{4a_{\text{in}}(\mu\delta+1)}c - \frac{(1-\xi)}{a_{\text{in}}}$$
$$- \frac{\xi}{2(\mu\delta+1)a_{\text{in}}},$$
$$\dot{\mu} = \frac{(\mu+1)}{w}\left(\frac{L}{a_{\text{in}}}w - \frac{\mu\xi c}{4a_{\text{in}}(\mu\delta+1)} - \frac{1-\xi}{a_{\text{in}}} - \frac{\xi}{2(\mu\delta+1)a_{\text{in}}}\right.$$
$$\left. - \frac{\xi\mu(1+\mu)c}{4a_{\text{im}}(\mu\delta+1)}\right),$$

with $\xi = \beta\sigma(1-\alpha)$ and $c = (4\delta(a_{\text{im}}/a_{\text{in}}) - 1)/(1 - a_{\text{im}}/a_{\text{in}})$. Linearizing these two differential equations and evaluating about the steady state, we get for the determinant of the Jacobian:

$$\operatorname{sign}\det(J) = \operatorname{sign}\left[\theta\left(-c(\mu^2\delta + 2\mu + 1) - c(a_{\text{im}}/a_{\text{in}}) + 2\delta(a_{\text{im}}/a_{\text{in}})\right)\right.$$
$$\left. + (L/a_{\text{in}})\left(1 - \delta - (\delta\mu^2 + 2\mu + 1)c\right)\right].$$

Given our assumptions the determinant is definitively negative. Hence, the eigenvalues are both real and of opposite sign, i.e., the steady state is (locally) stable in the saddle point sense.

From the above equations we get for the steady-state value of μ (with $\dot{w} = 0$, $\dot{\mu} = 0$) as the solution of

$$a_1\mu^2 + a_2\mu + a_3 = 0,$$

with

$$a_1 = c\left(L/(a_{\text{in}}\theta) + 1\right);$$
$$a_2 = (c-1)\left(L/(a_{\text{in}}\theta) + 1\right) + c\frac{a_{\text{im}}}{a_{\text{in}}} + \frac{1-\xi}{\xi}\frac{4a_{\text{im}}}{a_{\text{in}}}\delta$$

and

$$a_3 = \left(2a_{\text{im}}\theta\gamma - (1-\alpha)L\right)\left(a_{\text{in}}\theta(1-\alpha)\right)^{-1}.$$

Since $a_1 > 0$, then $a_3 < 0$ ensures for the case of a_2 being negative, that only one positive solution exists. For the case $a_2 > 0$, $a_3 < 0$ guarantees that there are not only negative solutions. Therefore, if $a_3 < 0$, (19) and (21) exactly intersect once in the first quadrant. The inequality $a_3 < 0$ holds if $L > L^* = 2a_{\text{im}}\theta\gamma(1-\alpha)^{-1}$.

Appendix B

Close to a steady-state with innovation but no imitation activities by defining $z = n^{m}/n$ we get $\dot{z} = (1 - z)g$. From (10), (15) and (17) we deduce

$$\dot{w} = \left(\theta + \frac{L}{a_{in}}\right) w - \frac{\xi\delta}{a_{in}(z\delta + 1 - z)} - \frac{1-\xi}{a_{in}} - \frac{\xi(1-z)}{2a_{in}(z\delta + 1 - z)},$$

$$\dot{z} = \frac{(1-z)}{a_{in}}\left(L - \frac{1}{w}\left(1 - \xi + \frac{\xi(1-z)}{2(z\delta + 1 - z)}\right)\right).$$

By linearizing and evaluating these equations about the steady-state ($z^{s} = 1$) we get a negative determinant of the Jacobian, i.e., the steady state is a saddle point.

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